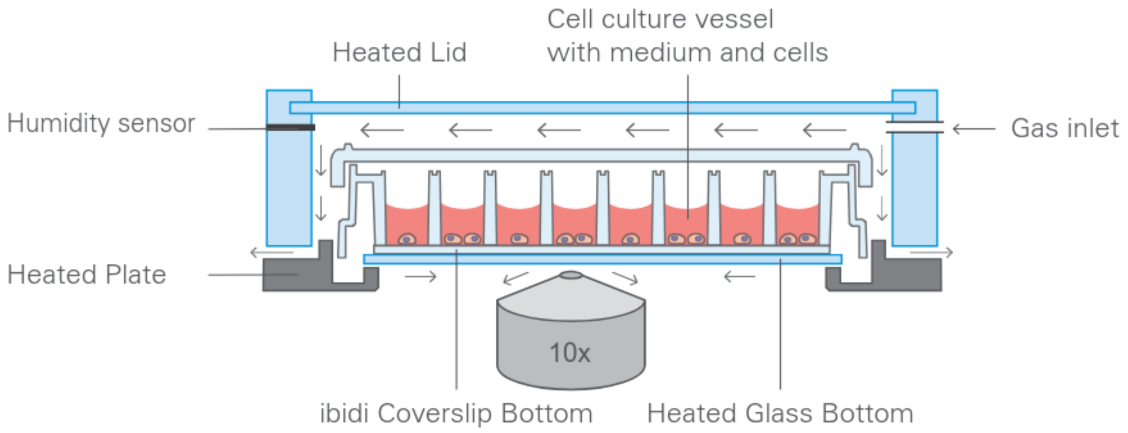
STAGE TOP BIO-CONTAINMENT WITH CONSIDERATION FOR A BSL-3 LAB

The Goal:
- Live-Cell Imaging of Microbes in BSL-3 Labs
- Low Cost Equipment
- Stage-Top Incubator Format

The Problem:
- Commercial Incubators not BSL-3 Containment viable
- Commercial Incubators not Suitable for Autoclave Sterilisation
- Commercial Incubators are Costly



THE PROJECT GOAL

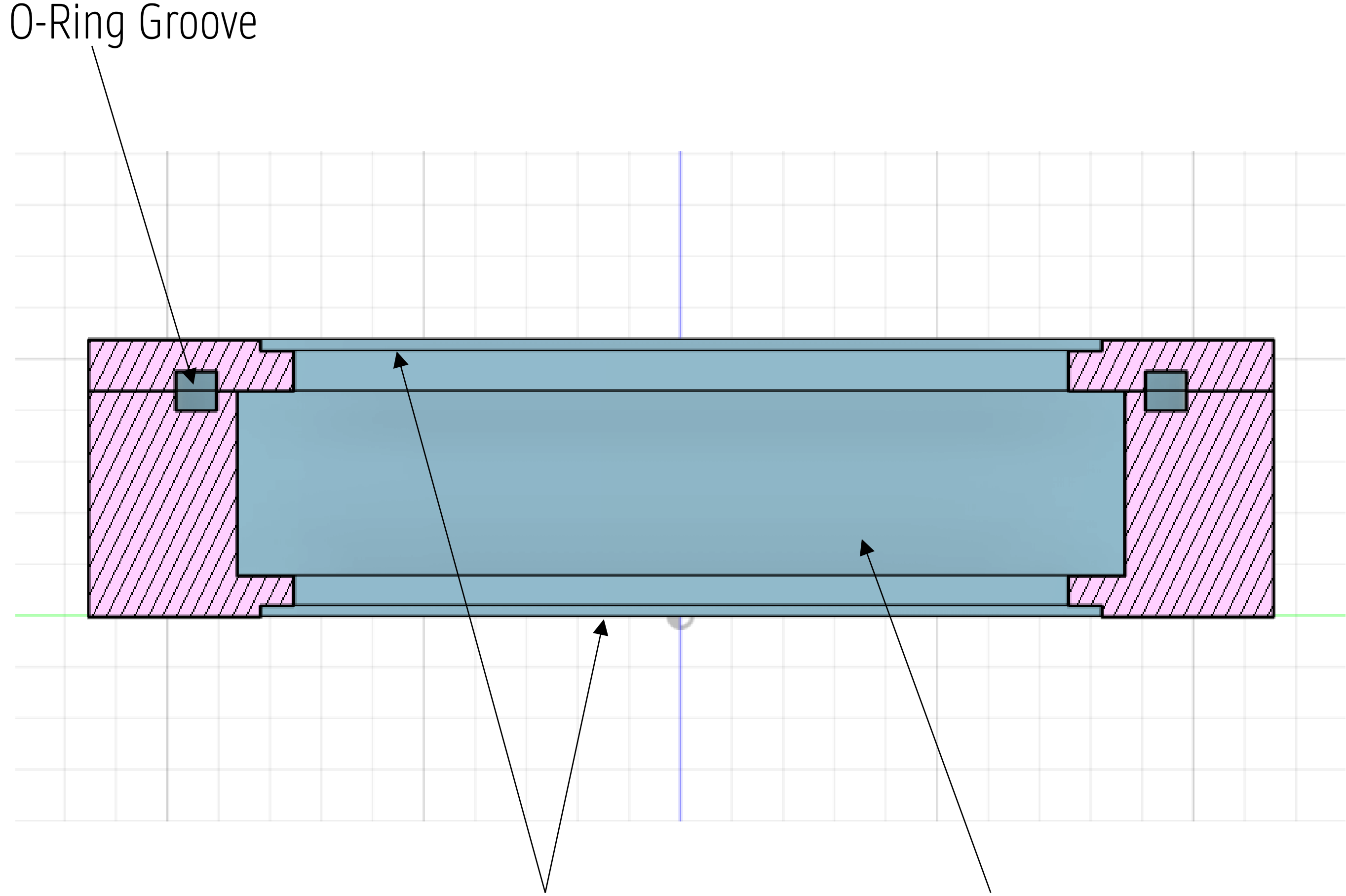
Immediate Goals

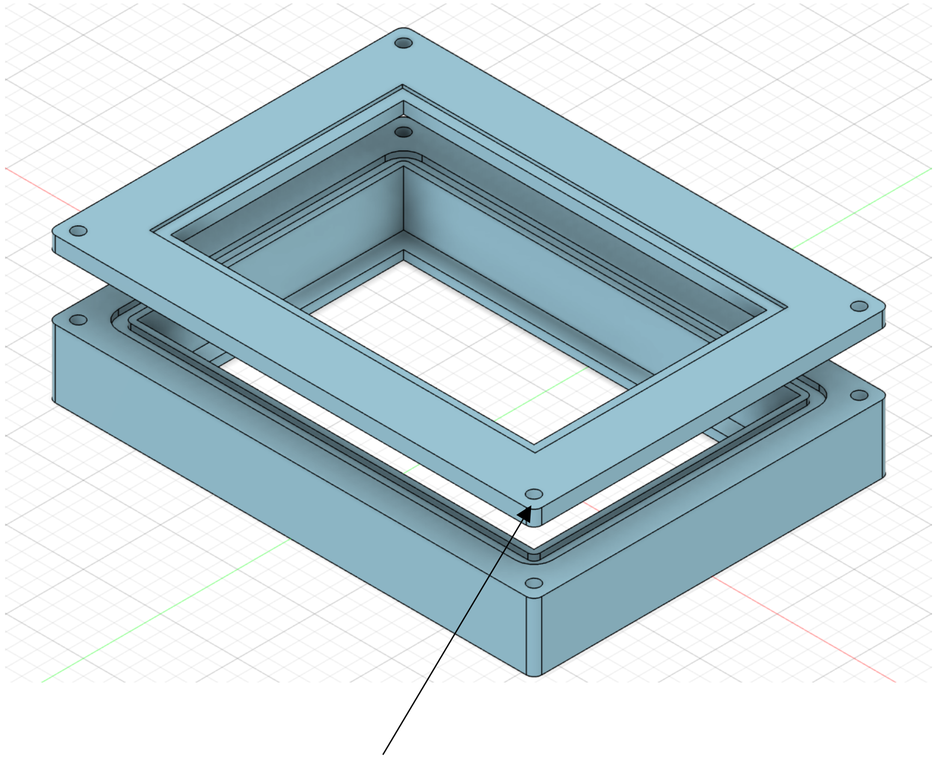
- Determine feasibility of designing a stage-top bio-containment system suitable for BSL-3
- Identify basic specifications required for the design of a BSL-3 Incubator

Future Goals

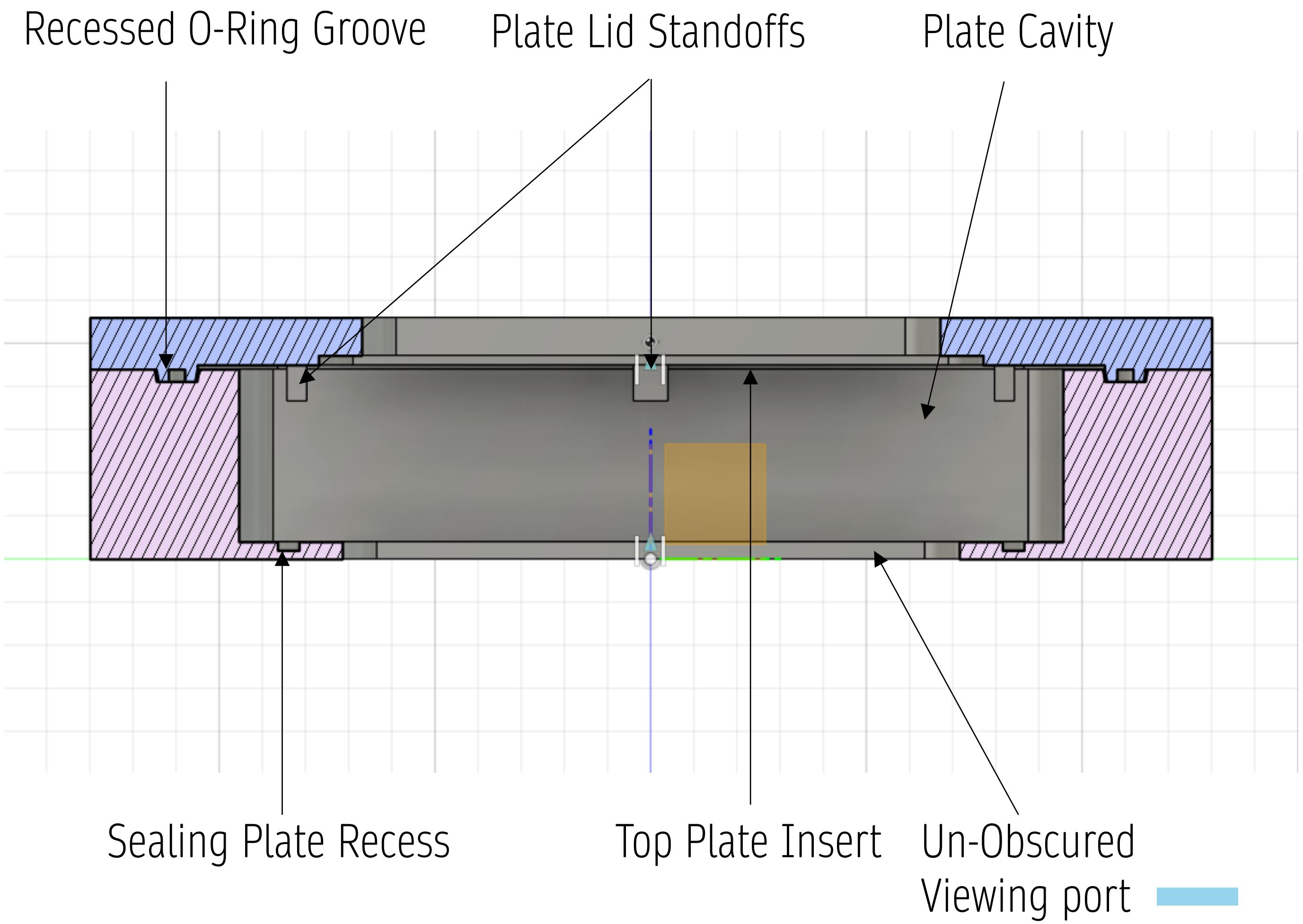
- Design and prototype a temperature controlled incubator that can operate in a BSL-3 lab.
- Determine feasibility of including CO2/O2 concentration control

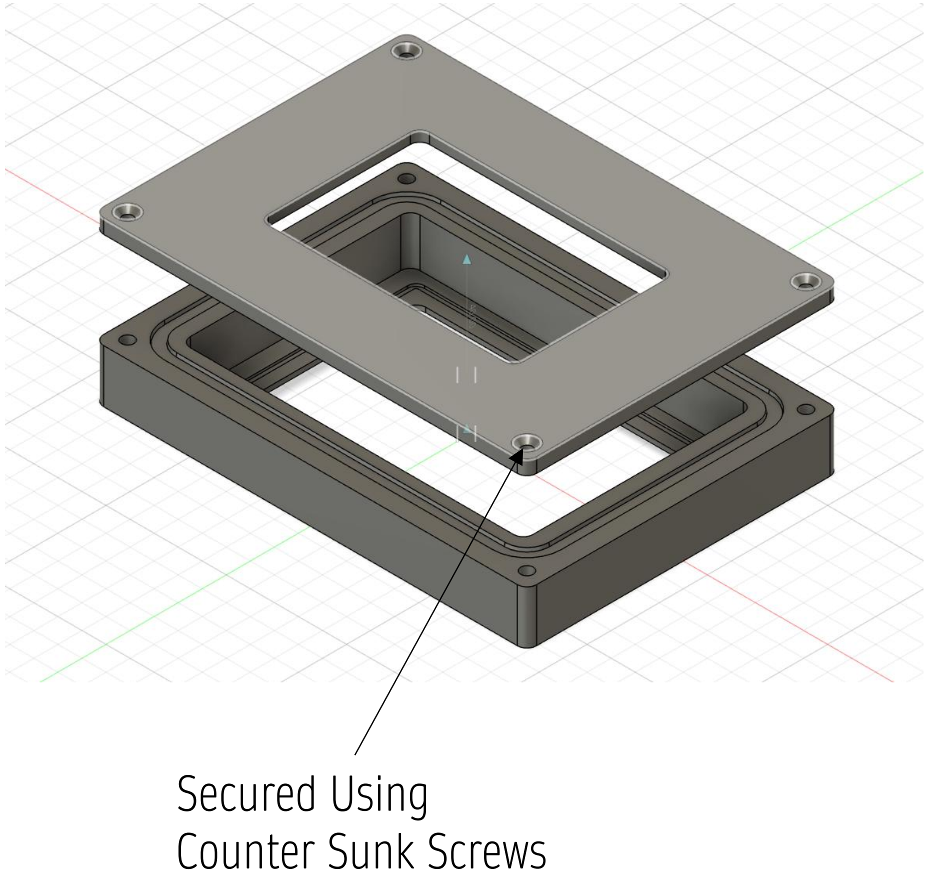
PROTOTYPING V1





PROTOTYPING V3.3





SUMMARY & FUTURE IMPROVEMENTS

- Silicone O-Ring Used to Seal Lid
- Silicone Gasket Used to Seal Base
- Compatible with Greiner Multi-well plates
- Secured using countersunk screws
- Unobstructed viewing from Top and Bottom

- Screw Mounting Tedious, and Supply Insufficient Mounting Pressure
- New Mechanism to Apply pressure to Multi-Well Plate
- Two Sealant Mechanisms Sub-Optimal
- Provisions for Gas and Temperature Control